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UNIFAC Parameter Table for Prediction of Liquid–Liquid Equilibria

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A UNIFAC group-interaction parameter table especially suited for prediction of liquid–liquid equilibria at temperatures between 10 and 40 °C has been developed. A total of 512 binary parameters representing the interactions between 32 different groups have been determined on the basis of approximately 100 binary and 300 ternary liquid–liquid equilibrium data sets. The parameters were estimated so that the reported mole fractions are represented as well as possible. The mean absolute deviation between experimental and predicted equilibrium composition is 2 mol %. Care was taken to reduce problems associated with false solutions to the liquid–liquid equilibrium criterion.

Introduction

Liquid–liquid equilibria (LLE) have in recent years gained increased interest in chemical technology. Due to the rising cost of energy, new separation processes based on extraction are becoming more attractive than before. Also, it may be feasible to operate known processes at new conditions, necessitating checks for liquid-phase stability at various points on the flow sheet.

Thus, the need for calculating and predicting LLE compositions has very much increased. In principle, LLE compositions may be calculated using any model for the excess Gibbs energy. Unfortunately—due to our lack of understanding of the behavior of liquids—it is not possible to quantitatively predict multicomponent LLE from binary data only or to predict LLE compositions using model parameters based on vapor–liquid equilibrium (VLE) data. This is shown in Table I.

If six UNIQUAC binary interaction parameters are fitted individually to each of the 17 ternary LLE data sets used as test mixtures by Sørensen et al. (1979a,b), the overall mean absolute deviation between experimental and calculated LLE compositions is 0.5 mol %. When the ternary LLE compositions are predicted using binary LLE and VLE data only, the deviation is 3.7 mol %. When the UNIFAC parameter table by Skjold-Jørgensen et al. (1979), which is based on VLE data is used, the deviation is 9.2 mol %. These predictions are clearly unsatisfactory compared to the individual fits.

Table I. Absolute Mean Deviation (Mole %) between Experimental and Calculated Mole Fractions for 17 Ternary LLE Test Systems (Sorensen et al., 1979)

UNIQUAC	individual fit to each of the 17 systems	0.48
UNIQUAC	"common parameters"	1.0
UNIQUAC	prediction from binaries	3.65
UNIFAC	LLE parameters	1.73
UNIFAC	VLE parameters	9.22

In order to predict multicomponent LLE as well as possible with the presently available models for the excess Gibbs energy, it is necessary to base the model parameters on binary and ternary LLE data. However, since the number of components of interest in LLE applications is very large, the data needed for obtaining these parameters are often not available.

The purpose of this article is to present a UNIFAC group-contribution parameter table especially suited for the prediction of LLE. We have used available LLE data (Sørensen and Arlt, 1979; see also Sørensen et al., 1979a,b) to calculate group-interaction parameters. These may be used to predict LLE compositions in mixtures for which no data are available, as long as the mixtures in question can be constructed from groups for which parameters are available. The UNIFAC model is that described in detail by Fredenslund et al. (1975, 1977) and by Skjold-Jørgensen et al. (1979). The same model is used in this work; only

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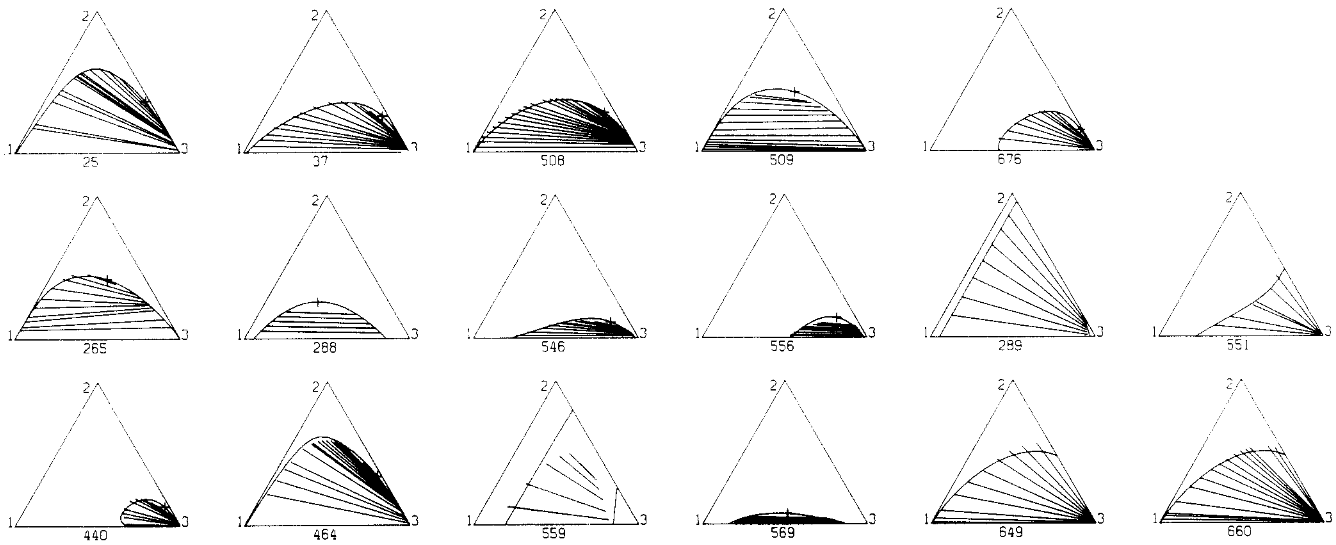

Figure 1. Experimental tie lines and predicted binodal curves for 17 test data sets; data set no., $\Delta\beta^\infty$, and components (1)–(2)–(3), respectively: 28, 0.2, 1,1,2-trichloroethane–2-propanone–water; 37, 2.1, diisopropyl ether–2-propanol–water; 265, 0.4, water–1-propanol–heptane; 288, 0.2, cyclohexane–benzene–furfural; 440, 8.2, phenol–2-propanol–water; 464, 0.6, trichloromethane–2-propanone–water; 508, 0.0, benzene–2-propanol–water; 509, 0.6, water–2-propanol–cyclohexane; 546, 0.1, furfural–2-propanol–water; 556, 0.3, 1-butanol–methanol–water; 559, 0.1, cyclohexane–cyclopentane–methanol; 569, 0.9, methanol–acetic acid, methyl ester–cyclohexane; 676, 3.4, 3-methyl 1-butanol–1-propanol–water; 289, 0.4, cyclohexane–2,2,4-trimethylpentane–water; 551, 0.1, furfural–2-methyl 1-propanol–water; 649, 2.0, benzene–2-methyl 1-propanol–water; 660, 7.0, benzene–1-butanol–water. $\Delta\beta^\infty = (\beta^\infty_{\text{exptl}} - \beta^\infty_{\text{pred}})/\beta^\infty_{\text{exptl}}$ where β^∞ is the ratio of x_2^{I} to x_2^{II} for $x_2 \to 0$. (I and II refer to phases I and II, exptl to experimental value, pred to prediction by UNIFAC.)

the parameter values differ. The equations are given in the Nomenclature section.

The existence of two different parameter tables for the same model may appear to be somewhat inconvenient. However, storing and using two different parameter tables should cause no difficulty. Only in few applications—for example, VLLE calculations—is it not so clear which parameters to use. In these cases, we suggest choosing the UNIFAC–VLE (Skjold-Jørgensen et al., 1979) parameters which are based on a broader data base, although we might expect rather poor results.

The loss of convenience is compensated by the encouraging results shown in Table I. Using the UNIFAC–LLE parameter table reported in this work, the mean absolute deviation between experimental and predicted equilibrium compositions is 1.7 mol %, i.e., five times less than for the UNIFAC–VLE parameter table and only four times worse than for individually fitted UNIQUAC parameters.

Figure 1 shows experimental tie lines and predicted binodal curves for the 17 test systems. The system numbers refer to the data bank of Sørensen and Arlt (1979). The relatively large deviation exhibited by UNIFAC for system number 559 (cyclohexane–cyclopentane–methanol) stems from the fact that the UNIFAC model cannot at the same time predict complete miscibility between methanol and cyclopentane and only partial miscibility between methanol and cyclohexane. When this system is not included in the calculation of the mean deviation, this number is reduced from 1.7 to 1.3 mol %.

Range of Applicability

UNIFAC–LLE group-interaction parameters are reported for 32 different groups representing hydrocarbons, water, alcohols, organic acids, halogenated hydrocarbons, nitriles, etc. The method does *not* apply to components with normal boiling points below 300 K, to strong electrolytes, or to polymers.

The temperature range is restricted to between 10 and 40 °C for two reasons. Firstly, 80% of the data in the LLE data bank lie within this region. Secondly, the temperature dependence of the model is not sufficiently adequate for permitting simultaneous correlation of LLE data over a wider temperature range.

It is difficult to ascertain the quality of the data used for establishing the parameters. There is no consistency test for LLE similar to that for VLE. Therefore, the choice of data to include in the parameter estimation is somewhat subjective.

The Parameter Table

Group volumes (R_k) and surface areas (Q_k) needed by UNIFAC are shown in Table II. The values of R_k and Q_k are determined in the same manner as by Fredenslund et al. (1975, 1977). UNIFAC–LLE parameters (a_{mn}) representing interactions between 32 different groups are shown in Table III. The majority of the parameters shown in Table III were estimated from binary and ternary experimental LLE data. Parameters marked with a superscript 1 for the 1-propanol and 2-propanol groups (P1 and P2, respectively) were estimated from artificial data sets, generated by describing the propanols by means of three alkane groups and one OH group.

The parameters marked with a superscript 2 were taken directly from the VLE parameter table of Skjold-Jørgensen et al. (1979). Most often this was done because no LLE data were available for the estimation of these parameters. It follows from Table I that if parameters marked 2 are used, the discrepancies between experimental and predicted LLE compositions may be expected on the average to be up to five times larger than if only parameters based on LLE data were used.

In a few cases, e.g., for the ACH/$ACCH_2$ group interaction, the VLE parameters of Skjold-Jørgensen et al. were chosen because the interactions in question proved to have only little effect on the calculated LLE compositions. It was thus advantageous to maintain these parameters at the values determined from VLE data.

Model Inadequacies

Some of the presently most used models for description of liquid phase nonidealities (UNIQUAC, NRTL, etc.) suffer from a serious computational disadvantage. In some cases the Gibbs energy curves described by these models show a much more complicated behavior than corre-

Table II. Group Volume and Surface Area Parameters

main group	sub group	no.	R_k	Q_k	sample group assignment
1 "CH_2"	CH_3	1	0.9011	0.848	butane: 2 CH_3, 2 CH_2
	CH_2	2	0.6744	0.540	
	CH	3	0.4469	0.228	2-methylpropane: 3 CH_3, 1 CH
	C	4	0.2195	0.000	2,2-dimethylpropane: 4 CH_3, 1 C
2 "C=C"	CH_2=CH	5	1.3454	1.176	1-hexene: 1 CH_3, 3 CH_2, 1 CH_2=CH
	CH=CH	6	1.1167	0.867	2-hexene: 2 CH_3, 2 CH_2, 1 CH=CH
	CH=C	7	0.8886	0.676	2-methyl-2-butene: 3 CH_3, 1 CH=C
	CH_2=C	8	1.1173	0.988	2-methyl-1-butene: 2 CH_3, 1 CH_2, 1 CH_2=C
3 "ACH"	ACH	9	0.5313	0.400	benzene: 6 ACH
	AC	10	0.3652	0.120	styrene: 1 CH_2=CH, 5 ACH, 1 AC
4 "$ACCH_2$"	$ACCH_3$	11	1.2663	0.968	toluene: 5 ACH, 1 $ACCH_3$
	$ACCH_2$	12	1.0396	0.660	ethylbenzene: 1 CH_3, 5 ACH, 1 $ACCH_2$
	ACCH	13	0.8121	0.348	cumene: 2 CH_3, 5 ACH, 1 ACCH
5 "OH"	OH	14	1.0000	1.200	2-butanol: 2 CH_3, 1 CH_2, 1 CH, 1 OH
6	P1	15	3.2499	3.128	1-propanol: 1 P1
7	P2	16	3.2491	3.124	2-propanol: 1 P2
8	H_2O	17	0.92	1.40	water: 1 H_2O
9	ACOH	18	0.8952	0.680	phenol: 5 ACH, 1 ACOH
10 "CH_2CO"	CH_3CO	19	1.6724	1.488	ketone group is 2nd carbon; 2-butanone: 1 CH_3, 1 CH_2, 1 CH_3CO
	CH_2CO	20	1.4457	1.180	ketone group is any other carbon: 3-pentanone: 2 CH_3, 1 CH_2, 1 CH_2CO
11	CHO	21	0.9980	0.948	acetaldehyde: 1 CH_3, 1 CHO
12	furfural	22	3.168	2.484	furfural: 1 furfural
13 "COOH"	COOH	23	1.3013	1.224	acetic acid: 1 CH_3, 1 COOH
	HCOOH	24	1.5280	1.532	formic acid: 1 HCOOH
14 "COOC"	CH_3COO	25	1.9031	1.728	butyl acetate: 1 CH_3, 3 CH_2, 1 CH_3COO
	CH_2COO	26	1.6764	1.420	butyl propanoate: 2 CH_3, 3 CH_2, 1 CH_2COO
15 "CH_2O"	CH_3O	27	1.1450	1.088	dimethyl ether: 1 CH_3, 1 CH_3O
	CH_2O	28	0.9183	0.780	diethyl ether: 2 CH_3, 1 CH_2, 1 CH_2O
	CH-O	29	0.6908	0.468	diisopropyl ether: 4 CH_3, 1 CH, 1 CH-O
	FCH_2O	30	0.9183	1.1	tetrahydrofuran: 3 CH_2, 1 FCH_2O
16 "CCl"	CH_2Cl	31	1.4654	1.264	1-chlorobutane: 1 CH_3, 2 CH_2, 1 CH_2Cl
	CHCl	32	1.2380	0.952	2-chloropropane: 2 CH_3, 1 CHCl
	CCl	33	1.0060	0.724	2-chloro-2-methylpropane: 3 CH_3, 1 CCl
17 "CCl_2"	CH_2Cl_2	34	2.2564	1.988	dichloromethane: 1 CH_2Cl_2
	$CHCl_2$	35	2.0606	1.684	1,1-dichloroethane: 1 CH_3, 1 $CHCl_2$
	CCl_2	36	1.8016	1.448	2,2-dichloropropane: 2 CH_3, 1 CCl_2
18 "CCl_3"	$CHCl_3$	37	2.8700	2.410	chloroform: 1 $CHCl_3$
	CCl_3	38	2.6401	2.184	1,1,1-trichloroethane: 1 CH_3, 1 CCl_3
19	CCl_4	39	3.3900	2.910	tetrachloromethane: 1 CCl_4
20	ACCl	40	1.1562	0.844	chlorobenzene: 5 ACH, 1 ACCl
21 "CCN"	CH_3CN	41	1.8701	1.724	acetonitrile: 1 CH_3CN
	CH_2CN	42	1.6434	1.416	propionitrile: 1 CH_3, 1 CH_2CN
22	$ACNH_2$	43	1.0600	0.816	aniline: 5 ACH, 1 $ACNH_2$
23 "CNO_2"	CH_3NO_2	44	2.0086	1.868	nitromethane: 1 CH_3NO_2
	CH_2NO_2	45	1.7818	1.560	1-nitropropane: 1 CH_3, 1 CH_2, 1 CH_2NO_2
	$CHNO_2$	46	1.5544	1.248	2-nitropropane: 2 CH_3, 1 $CHNO_2$
24	$ACNO_2$	47	1.4199	1.104	nitrobenzene: 5 ACH, 1 $ACNO_2$
25 "DOH"	$(CH_2OH)_2$	48	2.4088	2.248	1,2-ethanediol: 1 $(CH_2OH)_2$
26 "DEOH"	$(HOCH_2CH_2)_2O$	49	4.0013	3.568	diethylene glycol: 1 $(HOCH_2CH_2)_2O$
27 "pyridine"	C_5H_5N	50	2.9993	2.113	pyridine: 1 C_5H_5N
	C_5H_4N	51	2.8332	1.833	3-methylpyridine: 1 CH_3, 1 C_5H_4N
	C_5H_3N	52	2.667	1.553	2,3-dimethylpyridine: 2 CH_3, 1 C_5H_3N
28 "TCE"	CCl_2=CHCl	53	3.3092	2.860	trichloroethylene: 1 CCl_2=CHCl
29 "MFA"	$HCONHCH_3$	54	2.4317	2.192	methylformamide: 1 $HCONHCH_3$
30 "DMFA"	$HCON(CH_3)_2$	55	3.0856	2.736	dimethylformamide: 1 $HCON(CH_3)_2$
31 "TMS"	$(CH_2)_4SO_2$	56	4.0358	3.20	tetramethylenesulfone: 1 $(CH_2)_4SO_2$
32 "DMSO"	$(CH_3)_2SO$	57	2.8266	2.472	dimethyl sulfoxide: 1 $(CH_3)_2SO$

sponding to reality. UNIFAC also suffers from this disadvantage, since it is based on UNIQUAC.

The complicated contours of the Gibbs energy surfaces result in possibilities for multiple solutions to two-liquid phase calculations, and to false predictions of three-liquid phase equilibria (see Sørensen et al., 1979b, pp 57–59).

In the construction of the parameter table of this work, special care has been taken to reduce these problems. For ternary systems, three-liquid phase behavior gives rise to discontinuities in the binodal curve. Here the binodal curves were checked for continuity. In addition, the Gibbs energy surface was checked for convexity (indicating a minimum) at the end points of the calculated tie lines. These means are not sufficient for avoiding the above problems completely, but the problems have been reduced considerably. No false two-phase equilibria and only very few false three-phase equilibria have been observed for the correlations and predictions we have made until now. For the system xylene–acetonitrile–water, for example, we erroneously predict three-phase behavior in a limited concentration region. Of course, there is no absolute guarantee against false solutions in predictions of two-phase or three-phase equilibria.

Means of avoiding false solutions in a more stringent manner do exist. These consist of finding the global minimum when calculating equilibrium compositions during the parameter estimation, but this would be much too time consuming in our applications, and besides, it still

Table III. The UNIFAC Group-Interaction Parameter Table for Prediction of Liquid-Liquid Equilibria. [Superscript 1 = P1 or P2 Estimated from "Generated" Data. Superscript 2 = Parameter Taken from the UNIFAC Parameter Table of Skjold-Jorgensen et al. (1979)]

		1 CH_2	2 C=C	3 ACH	4 $ACCH_2$	5 OH	6 P1	7 P2	8 H_2O
1	CH_2	0	74.54	−114.8	−115.7	644.6	329.6	310.7	1300.0
2	C=C	292.3	0	340.7[2]	4102.0[2]	724.4	1731.0	1731.0	896.0
3	ACH	156.5	−94.78[2]	0	167.0[2]	703.9	511.5	577.3	859.4
4	$ACCH_2$	104.4	−269.7[2]	−146.8[2]	0	4000.0	136.6	906.8	5695.0
5	OH	328.2	470.7	−9.210	1.270	0	937.3	991.3	28.73
6	P1	−136.7	−135.7	−223.0	−162.6	−281.1	0	0	−61.29
7	P2	−131.9	−135.7	−252.0	−273.6	−268.8	0	0	5.890
8	H_2O	342.4	220.6	372.8	203.7	−122.4	247.0	104.9	0
9	ACOH	−159.8		−473.2	−470.4	−63.15	−547.2	−547.2	−595.9
10	CH_2CO	66.56	306.1	−78.31	−73.87	216.0	401.7[1]	−127.6[1]	634.8
11	CHO	146.1	517.0	−75.30	223.2	−431.3	643.4[1]	231.4[1]	623.7
12	FURF	14.78		−10.44	−184.9	444.7	−94.64	732.3	211.6
13	COOH	1744.0	−48.52	75.49	147.3	118.4	728.7[1]	349.1[1]	652.3
14	COOC	−320.1	485.6	114.8[2]	−170.0[2]	180.6	−76.64	−152.8	385.9
15	CH_2O	1571.0	76.44[2]	52.13[2]	65.69[2]	137.1	−218.1	−218.1	212.8
16	CCl	73.80	−24.36[2]	4.680[2]	122.9[2]	455.1	351.5	351.5	770.0
17	CCl_2	27.90	−52.71[2]			669.2	−186.1[1]	−401.6[1]	740.4
18	CCl_3	21.23	−185.1[2]	288.5[2]	33.61[2]	418.4	−465.7	−465.7	793.2
19	CCl_4	89.97	−293.70[2]	−4.70[2]	134.7[2]	713.5	−260.3	512.2	1205.0
20	ACCl	−59.06		777.8	−47.13	1989.0			390.7
21	CCN	29.08	34.78[2]	56.41	−53.29	2011.0			63.48
22	$ACNH_2$	175.8		−218.9	−15.41	529.0[2]			−239.8
23	CNO_2	94.34	375.4	113.6	−97.05	483.8	264.7	264.7	13.32
24	$ACNO_2$	193.6		7.180	−127.1	332.6			439.9
25	DOH	108.5		247.3	453.4	−289.3			−424.3
26	DEOH	81.49		−50.71	−30.28	−99.56			
27	PYR	−128.8		−225.3	−124.6	−319.2			203.0
28	TCE	147.3				837.9			1153.0
29	MFA	−11.91	176.7	−80.48					−311.0
30	DMFA	14.91	132.1	−17.78					−262.6
31	TMS	67.84	42.73	59.16	26.59				1.110
32	DMSO	36.42	60.82	29.77	55.97				

		9 ACOH	10 CH_2CO	11 CHO	12 FURF	13 COOH	14 COOC	15 CH_2O	16 CCl
1	CH_2	2255.0	472.6	158.1	383.0	139.4	972.4	662.1	42.14
2	C=C		343.7	−214.7		1647.0	−577.5	289.3[2]	99.61[2]
3	ACH	1649.0	593.7	362.3	31.14	461.8	6.0[2]	32.14[2]	−18.81[2]
4	$ACCH_2$	292.6	916.7	1218.0	715.6	339.1	5688.0[2]	213.1[2]	−114.1[2]
5	OH	−195.5	67.07	1409.0	−140.3	−104.0	195.6	262.5	62.05
6	P1	−153.2	−47.41[1]	−344.1[1]	299.3	244.4[1]	19.57	1970.0	−166.4
7	P2	−153.2	353.8[1]	−338.6[1]	−241.8	−57.98[1]	487.1	1970.0	−166.4
8	H_2O	344.5	−171.8	−349.9	66.95	−465.7	−6.320	64.42	315.9
9	ACOH	0	−825.7				−898.3		
10	CH_2CO	−568.0	0	−37.36[2]	120.3	1247.0	258.7	5.202[2]	1000.0
11	CHO		128.0[2]	0	1724.0	0.750	−245.8		751.8[2]
12	FURF		48.93	−311.6	0	1919.0	57.70		
13	COOH		−101.3	1051.0	−115.7	0	−117.6	−96.62	19.77
14	COOC	−337.3	58.84	1090.0	−46.13	1417.0	0	−235.7[2]	
15	CH_2O		52.38[2]			1402.0	461.3[2]	0	301.1[2]
16	CCl		483.9	−47.51[2]		337.1		225.4[2]	0
17	CCl_2		550.6		808.8	437.7	−132.9[2]	−197.7[2]	−21.35
18	CCl_3		342.2		203.1	370.4	176.5[2]	−20.93[2]	−157.1[2]
19	CCl_4	1616.0[2]	550.0		70.14	438.1	129.5[2]	113.9[2]	11.80[2]
20	ACCl		190.5			1349.0	−246.3[2]		
21	CCN		−349.2				2.410		
22	$ACNH_2$	−860.3	857.7			681.4			
23	CNO_2		377.0			152.4		−94.49[2]	
24	$ACNO_2$	−230.4	211.6						
25	DOH	523.0[2]	82.77		−75.23	−1707.0	29.86		
26	DEOH								
27	PYR	−222.7[2]			−201.9				
28	TCE		417.4		123.2	639.7			
29	MFA								
30	DMFA				−281.9				
31	TMS								
32	DMSO								

		17 CCl_2	18 CCl_3	19 CCl_4	20 ACCl	21 CCN	22 $ACNH_2$	23 CNO_2	24 $ACNO_2$
1	CH_2	−243.9	7.5	−5.550	924.8	696.8	902.2	556.7	575.7
2	C=C	337.1[2]	4583.0[2]	5831.0[2]		405.9[2]		425.7	
3	ACH		−231.9[2]	3000.0[2]	−878.1	29.13	1.640	−1.770	−11.19
4	$ACCH_2$		−12.14[2]	−141.3[2]	−107.3	1208.0	689.6	3629.0[2]	−175.6

Table III (*Continued*)

		17 CCl_2	18 CCl_3	19 CCl_4	20 ACCl	21 CCN	22 $ACNH_2$	23 CNO_2	24 $ACNO_2$
5	OH	272.2	−61.57	−41.75	−597.1	−189.3	−348.2[2]	−30.70	−159.0
6	P1	128.6[1]	1544.0	224.6				150.8	
7	P2	507.8[1]	1544.0	−207.0				150.8	
8	H_2O	370.7	356.8	502.9	−97.27	198.3	−109.8	1539.0	32.92
9	ACOH			4894.0[2]			−851.6		−16.13
10	CH_2CO	−301.0	12.01	−10.88	902.6	430.6	1010.0	400.0	−328.6
11	CHO								
12	FURF	−347.9	−249.3	61.59					
13	COOH	1670.0	48.15	43.83	874.3		942.2	446.3	
14	COOC	108.9[2]	−209.7[2]	54.57[2]	629.0[2]	−149.2			
15	CH_2O	137.8[2]	−154.3[2]	47.67[2]				95.18[2]	
16	CCl	110.5	249.2[2]	62.42[2]					
17	CCl_2	0		56.33[2]					
18	CCl_3		0	−30.10[2]		70.04[2]	−75.50		
19	CCl_4	17.97[2]	51.90[2]	0	475.8[2]	492.0[2]	1302.0[2]	490.9[2]	534.7[2]
20	ACCl			−255.4[2]	0	346.2		−154.5[2]	
21	CCN		−15.62[2]	−54.86[2]	−465.2	0			
22	$ACNH_2$		−216.3	8455.0[2]			0		179.9
23	CNO_2			−34.68[2]	794.4[2]			0	
24	$ACNO_2$			514.6[2]			175.8		0
25	DOH				−241.7		164.4[2]	481.3[2]	−246.0
26	DEOH								
27	PYR		−114.7[2]		−906.5	−169.7[2]	−944.9		
28	TCE								
29	MFA								
30	DMFA								
31	TMS								
32	DMSO								

		25 DOH	26 DEOH	27 PYR	28 TCE	29 MFA	30 DMFA	31 TMS	32 DMSO
1	CH_2	527.5	269.2	−300.0	−63.6	928.3	331.0	561.4	956.5
2	C=C					500.7	115.4	784.4	265.4
3	ACH	358.9	363.5	−578.2		364.2	−58.10	21.97	84.16
4	$ACCH_2$	337.7	1023.0	−390.7				238.0	132.2
5	OH	536.6	53.37	183.3	−44.44				
6	P1								
7	P2								
8	H_2O	−269.2		−873.6	1429.0	−364.2	−117.4	18.41	
9	ACOH	−538.6		−637.3[2]					
10	CH_2CO	211.6			148.0				
11	CHO								
12	FURF	−278.2		−208.4	−13.91		173.8		
13	COOH	572.7			−2.160				
14	COOC	343.1							
15	CH_2O								
16	CCl								
17	CCl_2								
18	CCl_3			18.98[2]					
19	CCl_4								
20	ACCl	124.8		−387.7					
21	CCN			134.3					
22	$ACNH_2$	125.3[2]		924.5					
23	CNO_2	139.8[2]							
24	$ACNO_2$	963.0							
25	DOH	0							
26	DEOH		0						
27	PYR			0					
28	TCE				0				
29	MFA					0			
30	DMFA						0		
31	TMS							0	
32	DMSO								0

would not be a guarantee against false solutions in predictions.

The "Propanol Problem"

All alcohols except the two propanols are described by means of CH_2 groups and the OH group. We were forced to treat the propanols separately, unless we were willing to accept large errors in the correlation of many systems with the propanols.

The ternary systems including propanol, especially the systems of water, propanol, and a hydrocarbon, show very complicated behavior and great variations through homologous series. The behavior is so complex that it is even necessary to distinguish between the two propanols and to define two independent propanol groups, P1 and P2 for 1-propanol and 2-propanol, respectively. This rather drastic step is justified by the relatively large number of ternary data sets involving the propanols. The explanation

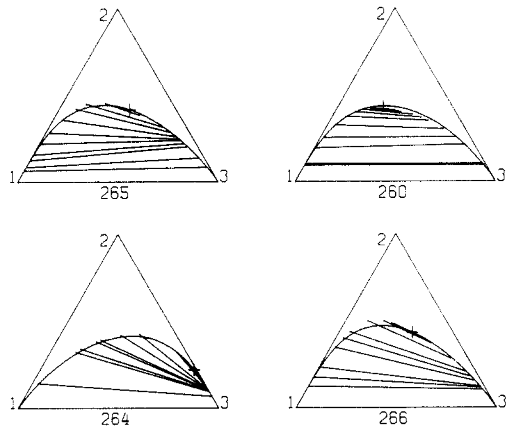

Figure 2. Typical systems of 1-propanol and 2-propanol with water and hydrocarbons; data set number and components (1)-(2)-(3), respectively: 260, water-2-propanol-hexane; 264, hexane-1-propanol-water; 265, water-1-propanol-heptane; 266, water-1-propanol-octane.

why the propanols show such complex behavior is not quite obvious.

Some examples of the complex propanol behavior are shown in Figure 2. Systems 264 (hexane-1-propanol-water), 265 (water-1-propanol-heptane), 266 (water-1-propanol-octane), and 260 (water-2-propanol-hexane) give an impression of both the large degree of variation through a homologous series, hexane/heptane/octane, the complex behavior of the individual systems, especially 264 and 265, and the large differences between identical systems with the two propanols (264 and 260).

Another aspect which make the propanol problem especially delicate is the problem of false three-phase equilibria. For systems like 264 and 265 the slopes of the tie lines change rapidly over a small distance on one of the branches of the binodal curves. Parameters estimated from such systems tend to result in predictions of three-phase equilibria unless special precautions are taken. This is done here by the previously described check for continuity of the predicted binodal curve and convexity of the calculated tie lines.

Parameter Estimation Method

We have used a least-squares technique for estimating the UNIFAC parameters from experimental LLE data. The objective function is

$$F(\tau) = \sum^{l}\sum^{k}\min\sum^{i}\sum^{j}[x_{ijkl} - \hat{x}_{ijkl}]^2 \quad (1)$$

$i = 1, 2, ..., N$; $j = 1,2$; $k = 1, 2, ..., M$; $l = 1, 2, ..., L$
(components) (phases) (tie lines) (data sets

x_{ijkl} is the experimental mole fraction, $\hat{x}_{ijkl}$ the calculated one. τ is the parameter vector. Binary and/or ternary data sets may be used in the parameter estimation. The program can cimultaneously treat a maximum of 10 binaries and 10 ternaries, 20 different components, and 9 different groups. A maximum of 12 parameters may be estimated in one run. The program does not need a good initial guess for the parameters. The CPU time requirements are low, mostly because we are using analytical derivatives, Michelsen's fast method for construction of the binodal curve (Fredenslund et al., 1980), and an effective optimization algorithm (Levenberg-Marquardt). A typical parameter estimation from 5 binary and 5 ternary systems with 10 different compounds and 7 different groups, in which 8 interactions are estimated, requires about 10-20 CPU s on the IBM 3033. For further details regarding the program, see Magnussen et al. (1980) and Sørensen et al. (1979).

By using the objective function (1) we have chosen to

Table IV. Prediction of Binary Mutual Solubilities at 25 °C Using the UNIFAC Group Contribution Model with Two Different Sets of Parameters. The Experimental Data Are Smoothed Values From Sorensen and Arlt (1979). Mole Fraction of Component *i* in Phase *j* Is Denoted x_{ij}. a_{mn} Is a Group-Interaction Parameter

component 2	x_{12}(exptl)	x_{21}(exptl)	x_{12}(calcd)	x_{21}(calcd)	x_{12}(calcd)	x_{21}(calcd)
water(1)-hydrocarbons(2)			a_{CH_2,H_2O} = 1310 K	a_{H_2O,CH_2} = 578 K	a_{CH_2,H_2O} = 1300 K	a_{H_2O,CH_2} = 342.4 K
pentane	4.40×10^{-4}	1.01×10^{-5}	5.20×10^{-4}	1.42×10^{-5}	7.05×10^{-4}	1.97×10^{-4}
octane	8.10×10^{-4}	1.10×10^{-7}	6.69×10^{-4}	9.13×10^{-8}	9.05×10^{-4}	4.53×10^{-6}
butane, 2-methyl-	4.52×10^{-4}	1.21×10^{-5}	5.18×10^{-4}	1.42×10^{-5}	7.02×10^{-4}	1.96×10^{-4}
pentane, 2,2,4-trimethyl-	8.50×10^{-4}	3.50×10^{-7}	7.01×10^{-4}	9.17×10^{-8}	9.49×10^{-4}	4.81×10^{-6}
hexane, 3-methyl-	4.12×10^{-4}	8.90×10^{-7}	6.16×10^{-4}	4.83×10^{-7}	8.34×10^{-4}	1.56×10^{-5}
cyclopentane	7.29×10^{-4}	4.01×10^{-5}	3.66×10^{-4}	4.45×10^{-5}	4.95×10^{-4}	3.81×10^{-4}
cyclohexane, methyl-	7.90×10^{-4}	2.57×10^{-6}	4.84×10^{-4}	1.44×10^{-6}	6.36×10^{-4}	2.86×10^{-5}
water(1)-alcohols(2)			a_{CH_2,H_2O} = 1310 K a_{OH,H_2O} = −264.7 K a_{OH,CH_2} = 299.6 K	a_{H_2O,CH_2} = 578 K $a_{H_2O,OH}$ = 149.0 K $a_{CH_2,OH}$ = 649.1 K	a_{CH_2,H_2O} = 1300 K a_{OH,H_2O} = 28.73 K $a_{CH_2,OH}$ = 328.2 K	a_{H_2O,CH_2} = 342.4 K $a_{H_2O,OH}$ = −122.4 K a_{OH,CH_2} = 644.6 K
1-butanol	0.512	1.92×10^{-2}	0.532	3.75×10^{-3}	0.534	1.85×10^{-2}
1-hexanol	0.290	1.04×10^{-3}	0.371	1.70×10^{-4}	0.351	1.60×10^{-3}
1-decanol	0.267	4.21×10^{-6}	0.247	3.87×10^{-7}	0.421	1.86×10^{-5}
2-pentanol	0.395	9.45×10^{-3}	0.434	7.78×10^{-4}	0.421	5.12×10^{-3}
1-propanol, 2,2-dimethyl-	0.309	7.36×10^{-3}	0.443	7.57×10^{-4}	0.433	5.38×10^{-3}
cyclohexanol	0.433	7.08×10^{-3}	0.380	4.89×10^{-4}	0.359	2.84×10^{-3}

Table V. Systems for a Typical Parameter Estimation. The Following Interactions Were Estimated: H_2O-CH_2, H_2O-OH, and OH-CH_2

systems	no. of tie lines
Binaries	
water-1-butanol	1
water-1-pentanol	1
water-1-hexanol	1
water-2-hexanol	1
water-2-hexanol, 2-methyl-	1
methanol-cyclohexane	1
methanol-hexane	1
Ternaries	
water-ethanol-hexane	8
water-ethanol-pentane, 2,2,4-trimethyl-	7
water-ethanol-nonane	6
1-butanol-methanol-water	9
1-butanol-ethanol-water	10

represent as well as possible the absolute mole fractions, i.e., to represent the binodal curves as well as possible. Thus, we have *not* emphasized the representation of small concentrations and, especially, the solute distribution ratios at small concentrations. The main reason for this is that UNIFAC is a generalized method. It does not specifically apply to one particular component, one particular system, or one particular type of systems; therefore it is not reasonable to emphasize certain regions of concentrations. As a consequence, the UNIFAC–LLE parameter table may be expected to yield reasonable estimates of the region of concentration where two liquid phases coexist, but not of the solute distribution ratios at low concentrations (see Figures 1 and 2).

It was shown in Table I that the absolute average deviation between predicted LLE compositions and experimental values is 1.7 mol % for 17 ternary systems. This value is typical for the several hundred data sets investigated in this work. Usually, the predicted solute distribution ratios are correct to within 50%. However, sometimes—especially at very low concentrations—the deviation is more than that.

Estimation of the Parameters

There are 32 groups in the parameter table. Theoretically, a total of 992 binary parameters are needed to describe the interactions between 32 groups. A simultaneous estimation of all these parameters would require too much computer time and space and, besides, involve insoluble weighting problems since the amounts of data for the different kinds of mixtures and substances vary so much. Thus, the parameters must be estimated in a suitable order, 2 or 4 or 6, etc., at a time. Of course, this gives rise to considerations regarding how to start and, next, how to continue the estimation.

By far the most important interaction of all is the one between water and the alkane group. Water is present in about 75% of the data sets in the data bank by Sørensen and Arlt (1979), and the alkane group is present in nearly every data set. So this interaction must be the first to be estimated, and great care should be taken regarding the types of data to be used for the parameter estimation.

Perhaps the most natural and straightforward way to estimate this interaction would be to use binary mutual solubility data for water and different alkanes since this kind of data is both abundant and quite accurate. This was done by Fredenslund et al. (1977) in the early versions of the UNIFAC parameter table.

The resulting parameters, a_{CH_2,H_2O} = 1310 K and a_{H_2O,CH_2} = 578 K, correlated and predicted the mutual solubilities reasonably well; see Table IV. But as more parameters were estimated, serious problems were encountered in the correlation and prediction of the solubility in water of some very important classes of substances. These substances are composed of alkane groups and a strong hydrophilic group such as the alcohols, esters, acids, ketones, etc. From Table IV it is evident that the predicted concentration of alcohols in water is about a factor of 10 too small. This

Table VI. Quaternary Predictions

system	no. of tie lines	components	av abs dev, mol % UNIFAC	UNIQUAC ("common parameters")
1	4	toluene–pyridine–water–ethanol	3.70	1.57
2	5	trichloromethane–propanone–water–formic acid	8.72	6.60
3	8	dichloromethane–butanoic acid–water–acetic acid	1.73	2.94
4	18	vinyl acetate–methyl acetate–water–methanol	5.53	5.38
5	37	3-buten-2-one–toluene–water–propanone	2.37	2.11
6	35	heptane–benzene–water–acetic acid, nitrile	2.83	1.84
7	24	heptane–benzene–water–sulfone, tetramethylene	2.26	4.09
8	9	trichloromethane–propanone–water–acetic acid	6.82	2.95
9	23	tetrachloromethane–benzene–water–acetic acid	1.59	0.89
10	22	hexane–methyl palmitate–acetonitrile–methyl oleate	9.44	0.97
11	8	hexane–benzene–water–formic acid, *N*-methyl amide	0.58	0.51
12	19	hexane–benzene–water–formic acid, *N,N*-dimethyl amide	1.54	3.40
13	6	1-hexene–benzene–water–formic acid, *N,N*-dimethyl amide	1.75	4.88
14	6	1-heptene–benzene–water–formic acid, *N,N*-dimethyl amide	1.79	7.41
15	5	cyclohexene–benzene–water–formic acid, *N,N*-dimethyl amide	1.96	1.73
16	6	1-hexene–benzene–water–formic acid, *N*-methyl amide	0.69	1.08
17	5	1-heptene–benzene–water–formic acid, *N*-methyl amide	0.63	0.54
18	20	2,2,4-trimethylpentane–cyclohexane–furfural–benzene	1.39	1.20
19	13	heptane–benzene–water–formic acid, *N*-methyl amide	0.59	1.92
20	17	heptane–benzene–water–formic acid, *N,N*-dimethyl amide (20 °C)	1.44	1.49
21	7	heptane–benzene–water–formic acid, *N,N*-dimethyl amide (40 °C)	0.80	1.56
23	22	1-butanol–propanone–water–ethanol	3.54	3.28
25	15	heptane–hexane–methanol–benzene	5.52	7.28
26	7	vinyl acetate–propanone–water–acetaldehyde	0.88	0.91
27	29	benzene–3-methylbutanoic acid, ethyl ester–water–ethanol	5.90	1.99
28	3	toluene–3-methylbutanoic acid, ethyl ester–water–ethanol	9.25	4.55
mean total deviation, mol %			3.09	2.80

Table VII. Interchange Energies ($a_{mn} + a_{nm}$) from VLE Based UNIFAC Parameters (Skjold-Jorgensen et al., 1979) and from the LLE Based Parameters from Table III. The Upper Values Are from VLE, the Lower Value from LLE

	CH_2	C=C	ACH	$ACCH_2$	OH	H_2O	CH_2CO
C=C	2320	0					
	367						
ACH	50	246	0				
	42	-					
$ACCH_2$	7	3832	20	0			
	−11	-	-				
OH	1143	9388	726	829	0		
	973	1195	695	4001			
H_2O	1618	1327	1266	6073	124	0	
	1642	1117	1232	5899	−94		
ACOH	3100	-	3440	6971	−247	−99	
	2095	-	1176	−178	−259	−251	
CH_2CO	503	442	166	314	249	277	0
	539	650	515	843	283	463	
CHO	1183	-	-	-	37	−25	91
	304	302	287	1441	978	274	-
COOC	347	-	92	5518	347	10014	159
	652	−92	-	-	376	380	318
CH_2O	335	366	84	279	266	226	58
	2233	-	-	-	400	277	-
$ACNH_2$	6584	-	1319	1745	181	−127	-
	1078	-	−217	674	-	−350	1868
CCN	622	441	190	5958	192	355	194
	726	-	86	1155	1822	262	81
COOH	979	1080	600	872	48	−80	372
	1883	1598	537	486	14	187	1146
CCl	127	75	−14	9	638	1024	95
	116	-	-	-	517	1086	1484
CCl_2	88	284	-	-	636	1079	139
	−216	-	-	-	941	1111	250
CCl_3	62	4398	57	21	644	1181	198
	29	-	-	-	357	1150	354
CCl_4	26	5537	−2	−7	999	1699	333
	84	-	-	-	672	1708	539
ACCl	180	-	301	249	535	1599	-
	866	-	−100	−154	1392	293	1093
CNO_2	629	492	178	3532	403	639	−5
	651	801	112	-	453	1552	777
$ACNO_2$	6084	-	2020	-	-	760	-
	769	-	−4	−303	174	473	−117
Furf	329	-	93	384	401	211	154
	398	-	21	531	304	279	169
DOH	3165	-	432	5126	−51	-	-
	636	-	606	791	247	−694	294

trend also prevails for ketones, acids, esters, etc.

Another way to establish the extremely important interaction between water and the alkane group is to use data, both binary and ternary, for mixtures of water with substances such as alcohols, ketones, esters, and alkanes, all of which contain the alkane group. When this is done, the resulting parameters are a_{CH_2,H_2O} = 1300 K and a_{H_2O,CH_2} = 342.4. These parameters do not predict the mutual solubilities of water and alkanes as well as the previous values, as seen in Table IV. The very small concentrations of alkanes in water are now a factor of 10–50 too large, but from Table IV, it is also clear that the concentrations of alcohols in water are now much better than before. This is analogous for ketones, esters, acids, etc. Also, the correlation and prediction of many ternary systems of water, alkanes, and alkane group-containing substances were substantially improved.

We decided to use the latter procedure for establishing the water–alkane interaction since we deemed it more important to represent reasonably well the mutual solubilities of water and alcohols, ketones, acids, esters, ethers, etc., than the water–alkane solubilities.

In the actual estimation of the H_2O–CH_2 interaction parameters, we used 7 binary and 5 ternary data sets of water–alcohol, methanol–alkane, water–alcohol–alkane, and water–alcohol–alcohol mixtures, as seen in Table V. In this case a total of 6 interaction parameters were estimated simultaneously: those for H_2O–CH_2, H_2O–OH and OH–CH_2. The number of data sets involved is typical for the parameter estimations performed in this work.

The order in which we estimated the rest of the parameters was determined mainly by the order of importance of the groups, that is, how often they appear in the published experimental data sets.

Quaternary Predictions

Table VI shows the absolute mean deviations between experimental tie lines and those predicted from UNIFAC for 26 different quaternary systems. The average deviation for the 26 systems is 3.1 mol %. When the UNIQUAC "common" parameters by Sørensen and Arlt (1979) are used to predict the tie lines, the average deviation is 2.8 mol %. "Common" parameters means that, e.g., the A–B interaction parameters for the system ABC are the same as for the systems ABD, ABE, etc. While the predicted equilibrium concentrations are of similar quality, the UNIQUAC "common" parameters in general yield much better predictions of solute distribution ratios at small concentrations than does UNIFAC. Thus, in those cases where UNIQUAC "common" parameters are available, they should be used in the prediction of LLE compositions rather than UNIFAC.

In the quaternary predictions, the calculated mole fractions are obtained from a flash calculation originating at the midpoint of each experimental tie line. Thus—unlike for ternary systems—the predicted tie lines are not as close as possible to the experimental ones.

LLE and VLE Parameters

A comparison between the VLE based UNIFAC parameters (Skjold-Jørgensen et al., 1979) and the LLE based parameters presented in Table III does not reveal many similarities between the parameters. This is not surprising since the group-interaction parameters are known to be correlated.

If the UNIFAC model was a perfect model, i.e. (1) if the solution of groups concept was not an approximation and (2) if the model described the group-interactions correctly, the interchange energy between two groups should be characteristic for a given pair of groups. It should hence not be dependent on the basis from which it is derived, i.e., VLE or LLE.

The interchange energy, w, between groups m and n can be calculated from a summation of the group-interaction parameters a_{mn} and a_{nm} as can be seen from eq 2 and 3.

$$a_{mn} = (u_{mn} - u_{nn})/R; \; a_{nm} = (u_{nm} - u_{mm})/R \qquad (2)$$

$$a_{mn} + a_{nm} = [u_{mn} - \tfrac{1}{2}(u_{mm} + u_{nn})]\cdot\frac{2}{R} = w\cdot\frac{2}{R} \qquad (3)$$

u_{mn} represents the interaction energy between groups m and n. R is the universal gas constant.

Table VII shows some rounded values of $a_{mn} + a_{nm}$. From the 90 pairs of values which allow comparisons it can be seen that: (1) about 50% of the pairs of values are of similar size and about half of these are almost identical; (2) for many of the remaining pairs the values are of the same magnitude and only for about 10% of the pairs one finds widely different values (e.g., $COOC/H_2O$ and $C{=}C/CH_2$).

Conclusions

The parameter table presented in this work allows semiquantitative predictions of liquid–liquid equilibria by means of the UNIFAC group contribution method.

The parameters were estimated so as to represent the absolute mole fractions the best way possible. In ternary systems, the binodal curve is usually predicted very well, but small concentrations may be afflicted with large relative errors. Hence, the predicted solute distribution ratios in the dilute regions may be quite erroneous.

The parameters were fitted to experimental LLE data at temperatures between 10 and 40 °C. No provisions regarding extrapolation with respect to temperature have been made. To provide for this possibility in a group-contribution model one needs both an improved temperature dependence of the model and more LLE data outside the above temperature range.

In the parameter estimation, special care was taken to reduce the problems of false solutions. However, these problems can not be totally prevented.

The reason for making a parameter table especially suited for LLE, and not a table which could be used simultaneously for both VLE and LLE, is that the present form of UNIFAC does not allow simultaneous correlation. To do so, model refinements need to be made, especially concerning its temperature dependence. Such work is in progress.

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Nomenclature

$\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R$
$\ln \gamma_i^C = (\ln \phi_i/x_i + 1 - \phi_i/x_i) - {}^1/_2 z q_i(\ln \phi_i/\theta_i + 1 - \phi_i/\theta_i)$
$\phi_i = x_i r_i / \sum_j x_j r_j$
$\theta_i = x_i q_i / \sum_j x_j q_j$
$\sum_j$: summation over all components
$r_i = \sum_k \nu_{ki} R_k$
$q_i = \sum_k \nu_{ki} Q_k$
$\sum_k$: summation over all groups
R_k = volume parameter for group k
Q_k = surface area parameter for group k
ν_{ki} = number of groups of type k in molecule i
x_i = liquid mole fraction of component i
z = coordination number = 10
$\ln \gamma_i^R = \sum_k \nu_{ki}(\ln \Gamma_k - \ln \Gamma_k^{(i)})$
$\ln \Gamma_k = Q_k[1 - \ln(\sum_m \theta_m \psi_{mk}) - \sum_m(\theta_m \psi_{km}/\sum_n \theta_n \psi_{nm})]$
$\sum_k$, $\sum_m$, and $\sum_n$: summation over all groups
$\psi_{nm} = \exp(-a_{nm}/T)$
$\theta_m = Q_m X_m / \sum_n Q_n X_n$
$X_m = \sum_j \nu_{mj} x_j / \sum_j \sum_n \nu_{nj} x_j$
a_{mn} = group interaction parameter for the interaction between groups m and n

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